Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

1. Problem Statement

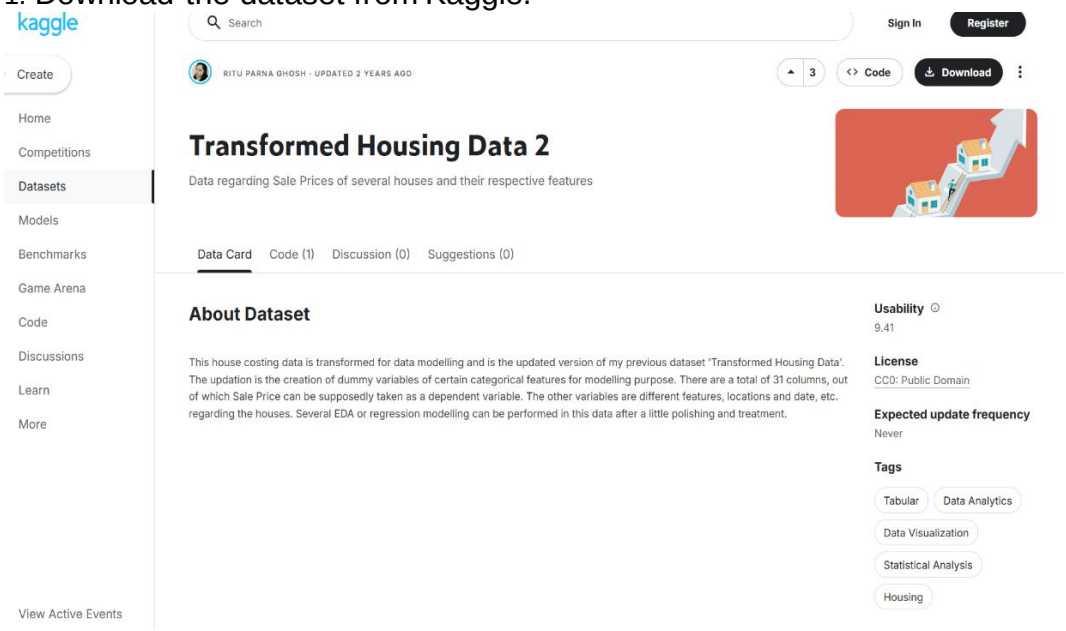
The objective of this project is to analyze housing market trends using Tableau. The analysis focuses on understanding how house sale prices are influenced by features such as house age, renovation status, number of bedrooms, bathrooms, floors, and basement area. Using data visualization techniques, the project aims to uncover patterns and insights that help explain variations in housing prices.

2. Data Collection

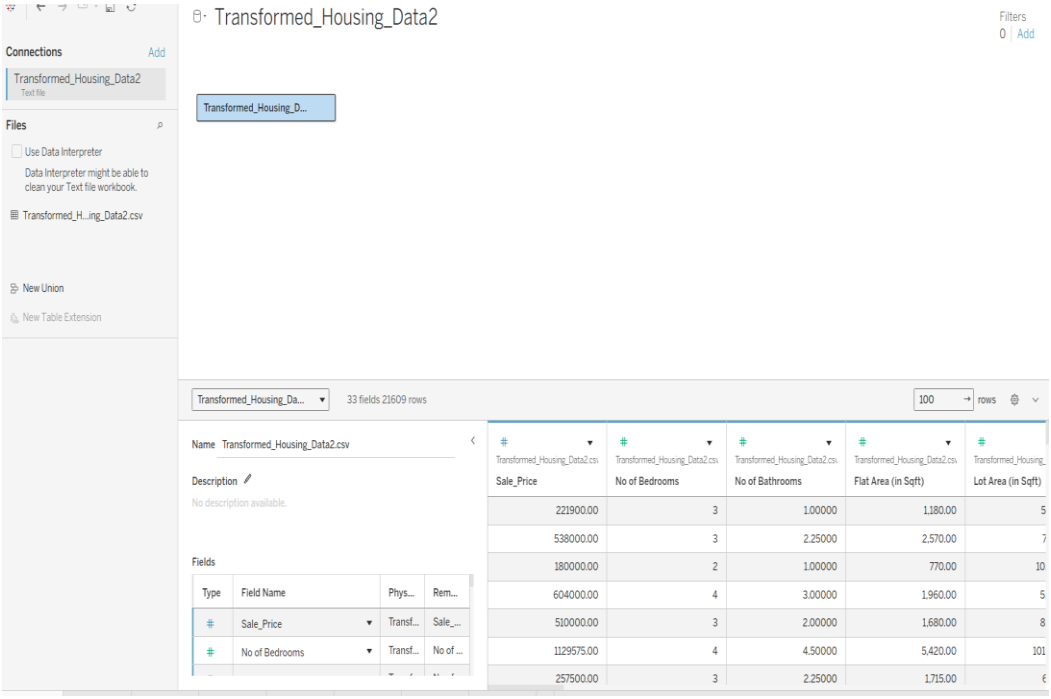
The dataset for this project was collected from an online source such as Kaggle. The dataset contains housing information including sale price, renovation status, number of bedrooms, bathrooms, floors, and house age.

Steps followed:

1. Download the dataset from Kaggle.



2. Saved the dataset in CSV format.
3. Imported the dataset into Tableau for analysis.



3. Data Understanding

Before visualization, the dataset was reviewed to understand the available columns and data types.

Important fields in the dataset:
- Sale Price
- Basement Area (Sqft)
- Years Since Renovation
- Number of Bedrooms
- Number of Bathrooms
- Number of Floors
- House Age

Understanding the dataset helped identify how each variable contributes to housing prices.

4. Data Cleaning and Preparation

To ensure accurate analysis, the dataset was cleaned and prepared before visualization.

Steps performed:
- Checked for missing values

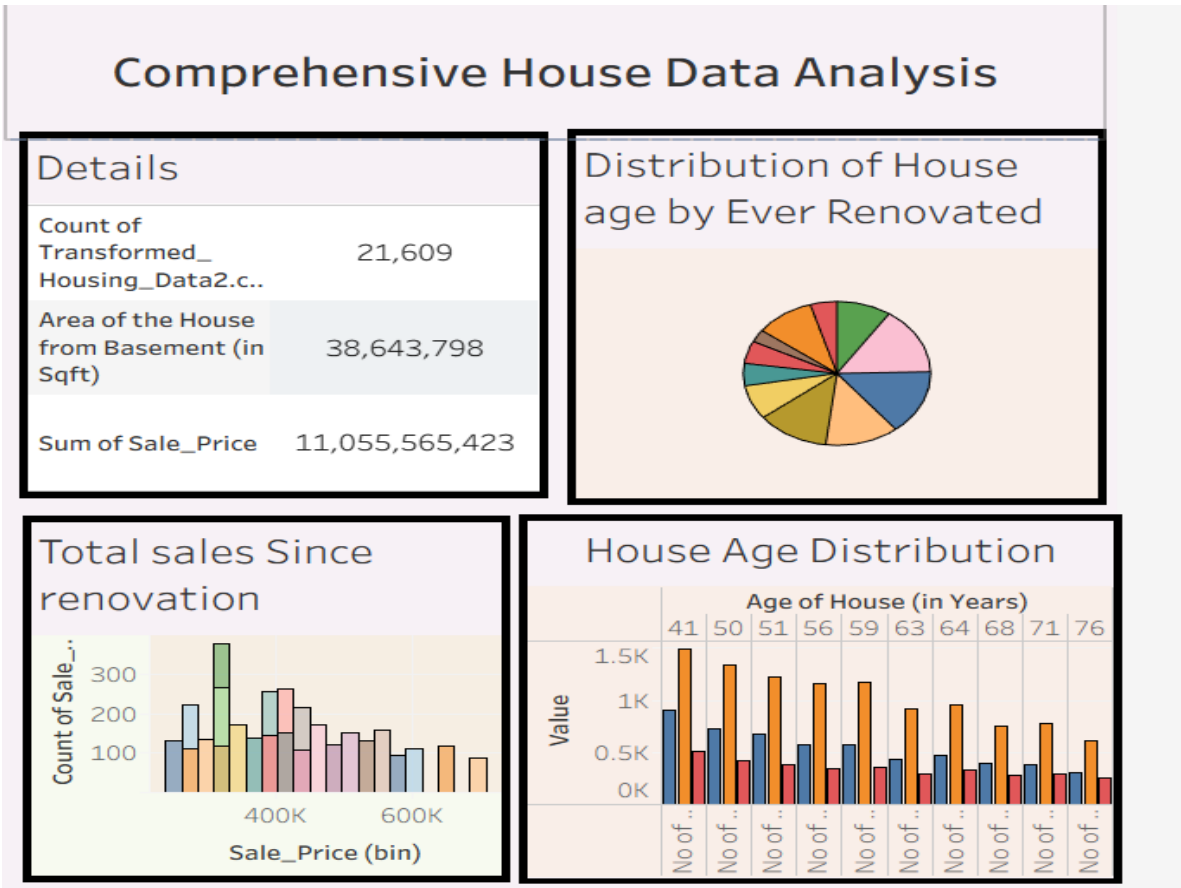
• Verified correct data types (numeric, categorical, etc.)
• Renamed fields for better understanding
• Created calculated fields such as house age and years since renovation
• Validated the dataset for consistency

5. Tableau Visualizations

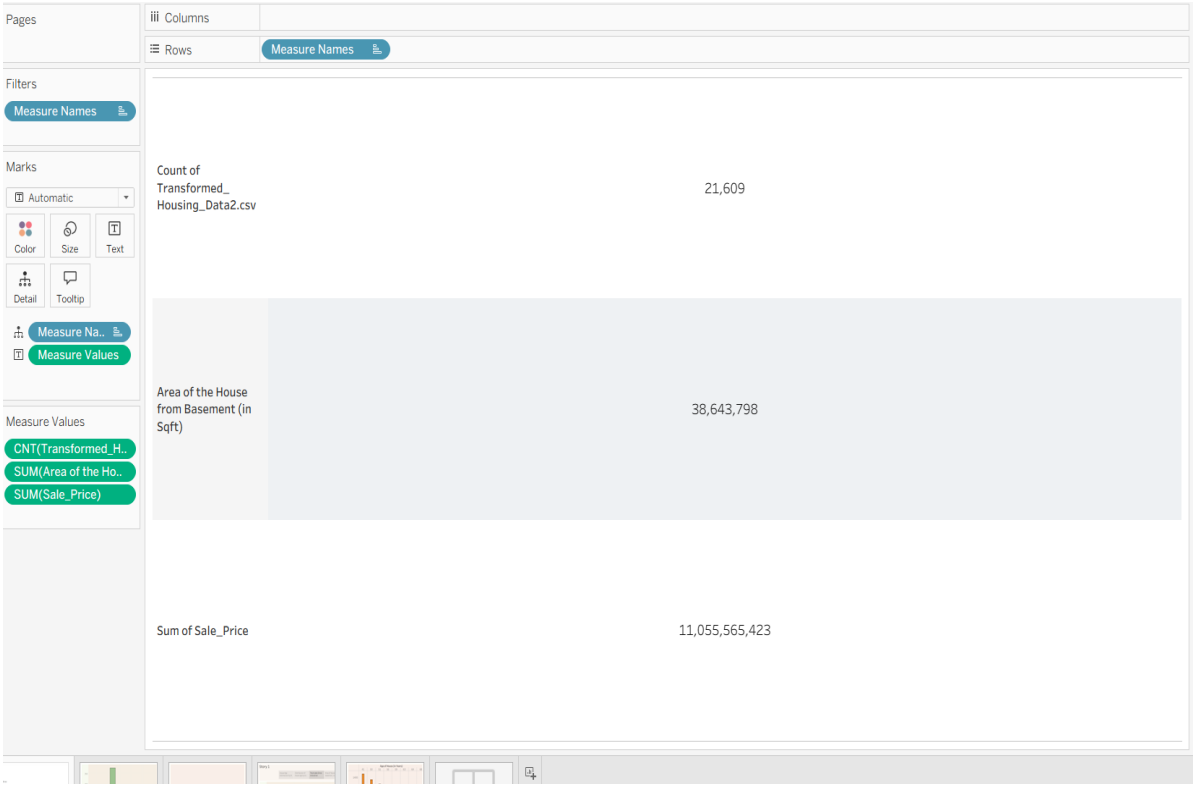
Several visualizations were created in Tableau to analyze the housing market.

Visualizations include:
• Count of Transformed Housing Data
• Average Sale Prices
• Area of House from Basement (in Sqft)
• Total Sales by Years Since Renovation
• Distribution of House Age by Renovation Status
• House Age Distribution by Number of Bathrooms, Bedrooms, and Floors

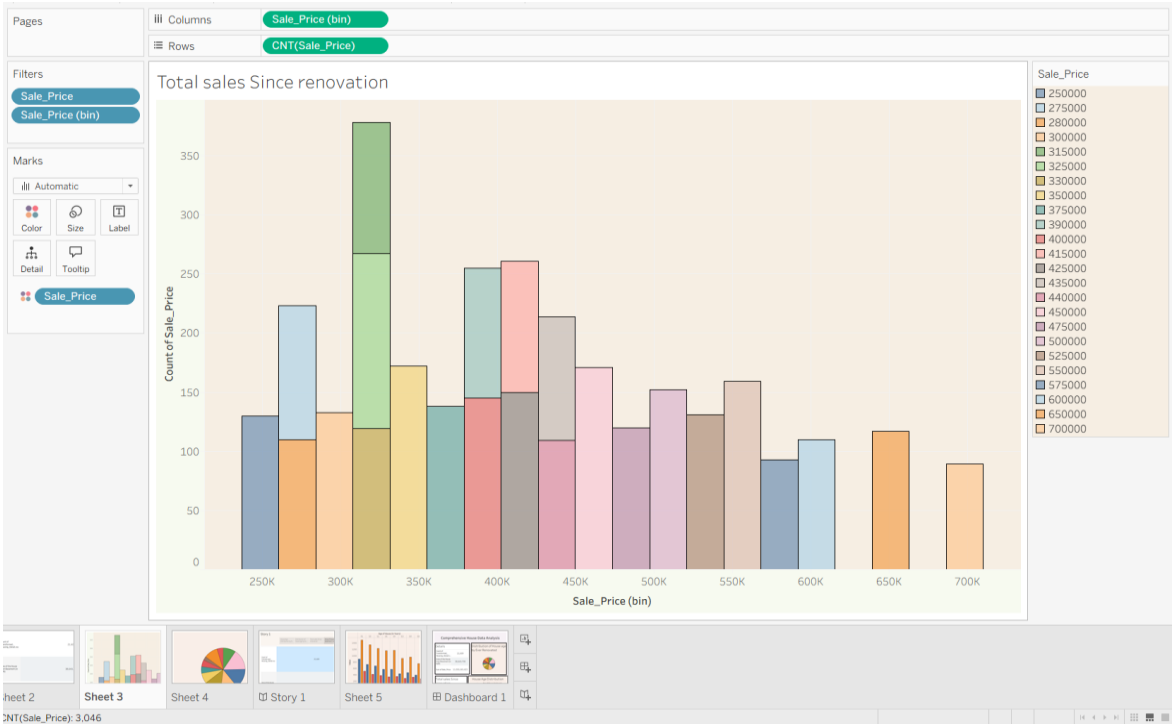
Insert Screenshot 1 – Tableau Dashboard Overview



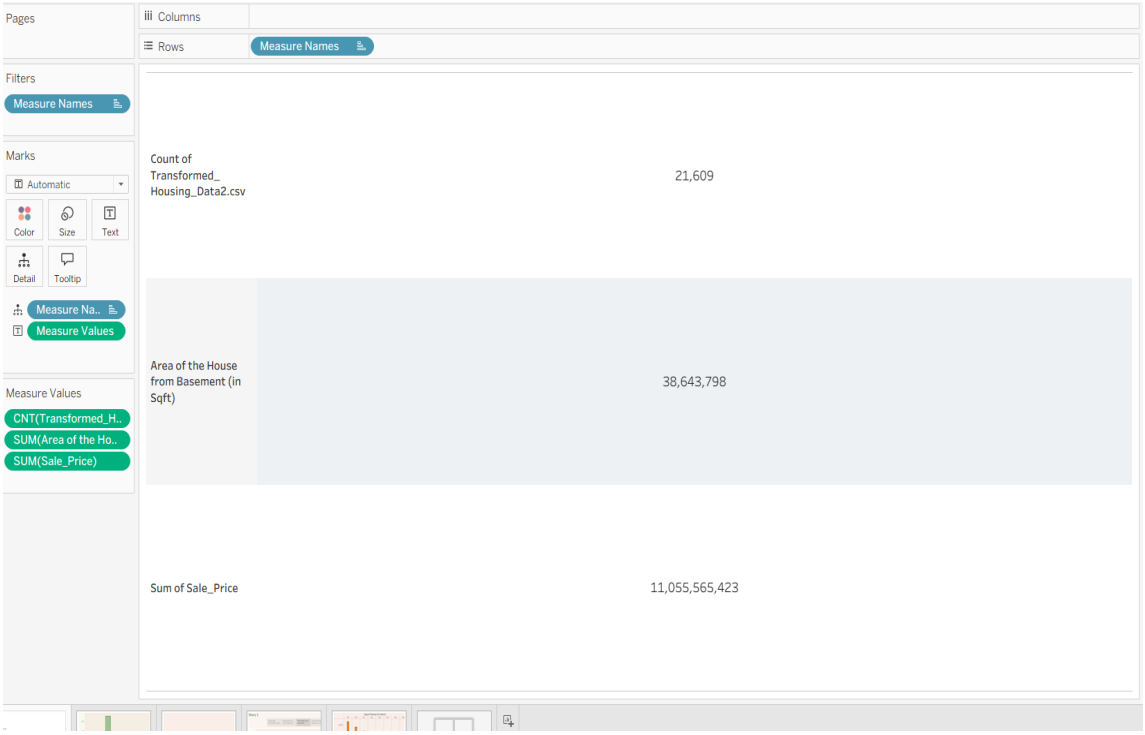
Insert Screenshot 2 – Count of Transformed Housing Data



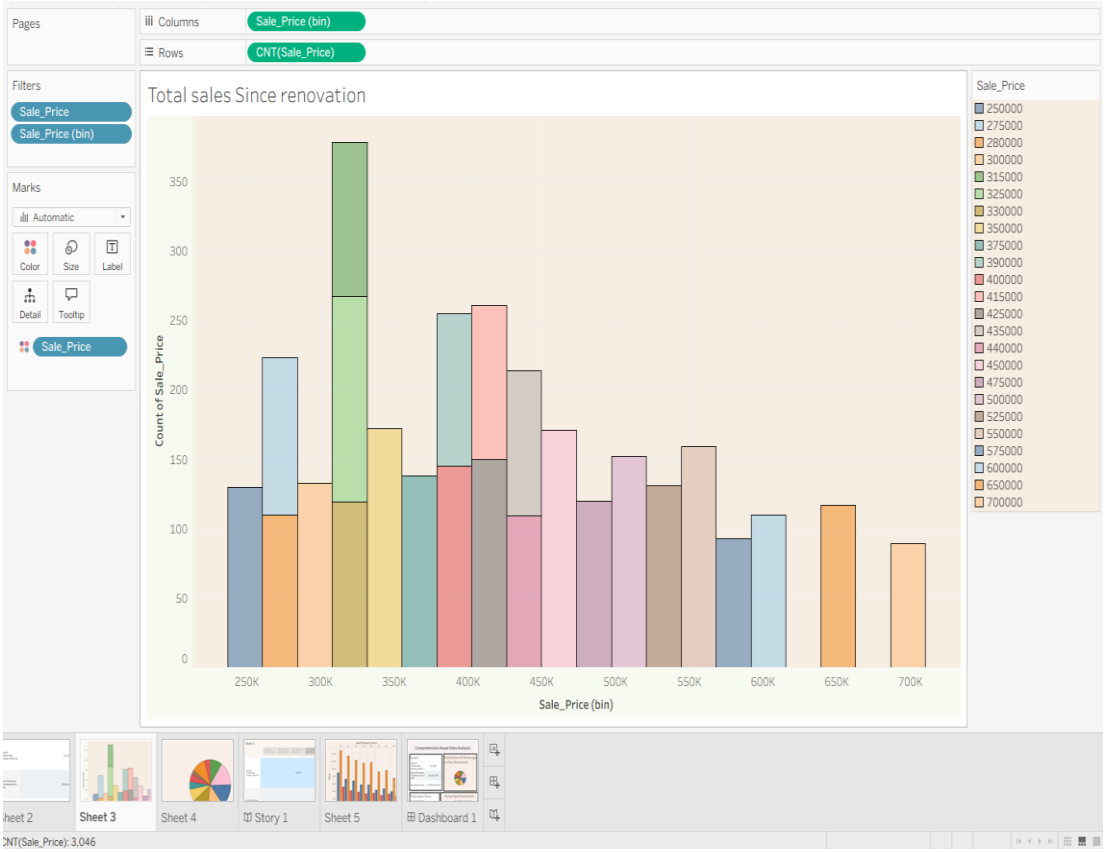
Insert Screenshot 3 – Average Sale Prices



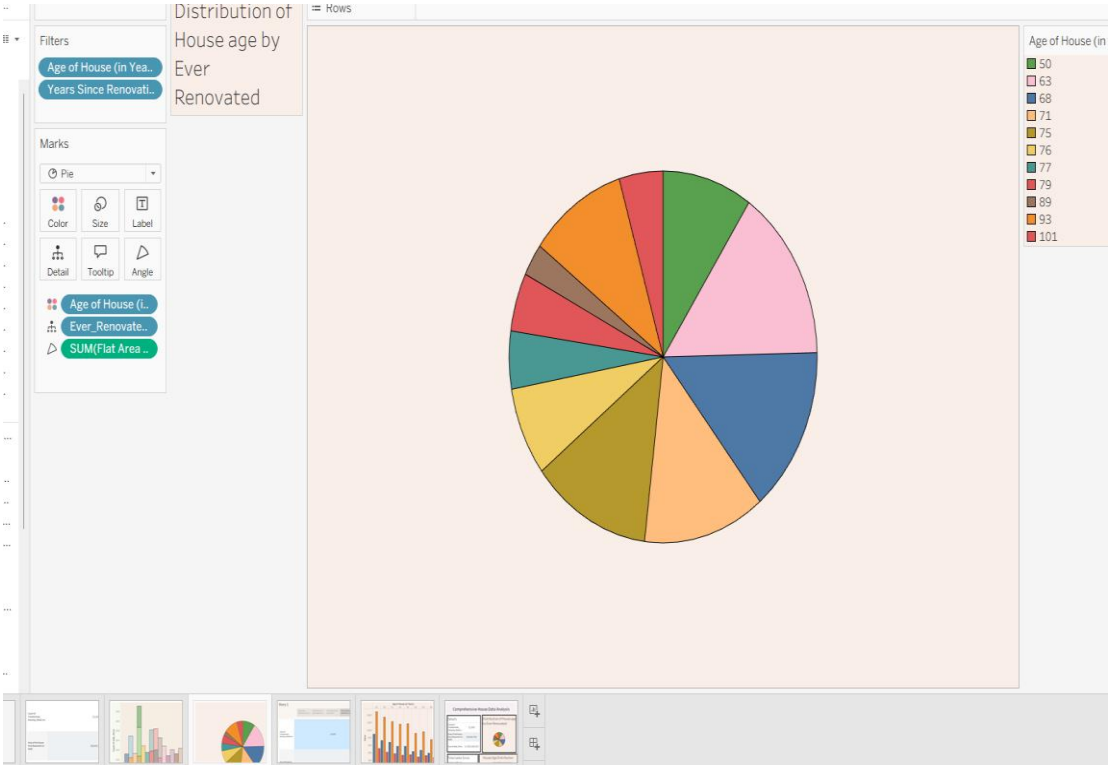
Insert Screenshot 4 – Area of House from Basement



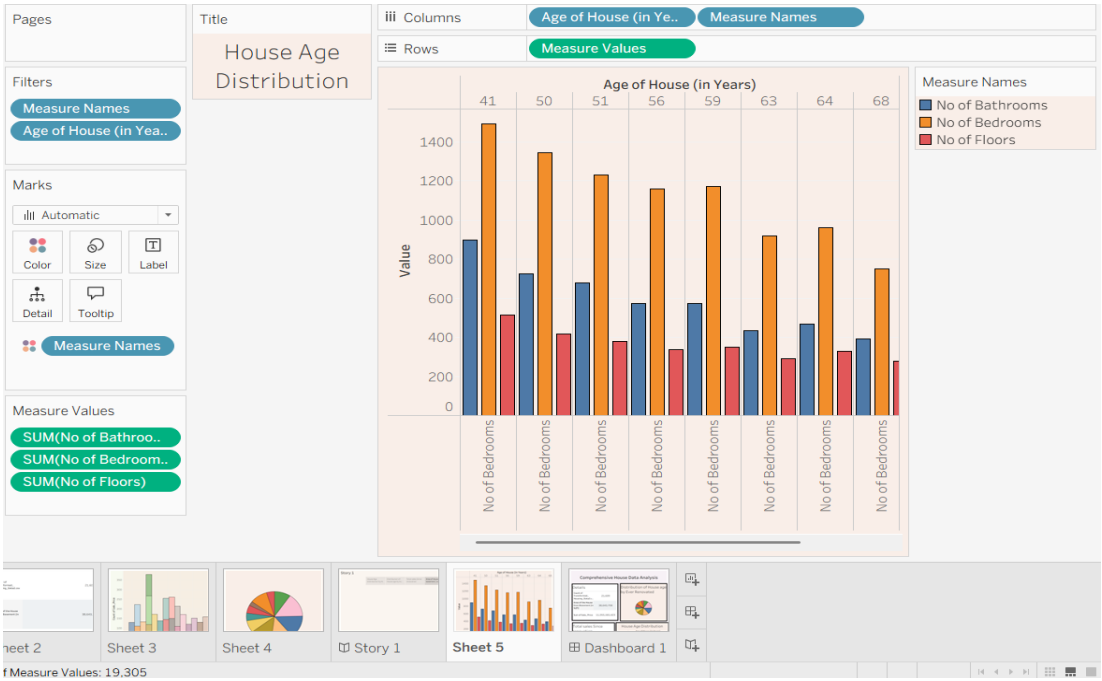
Insert Screenshot 5 – Total Sales by Years Since Renovation



Insert Screenshot 6 – Distribution of House Age by Renovation Status

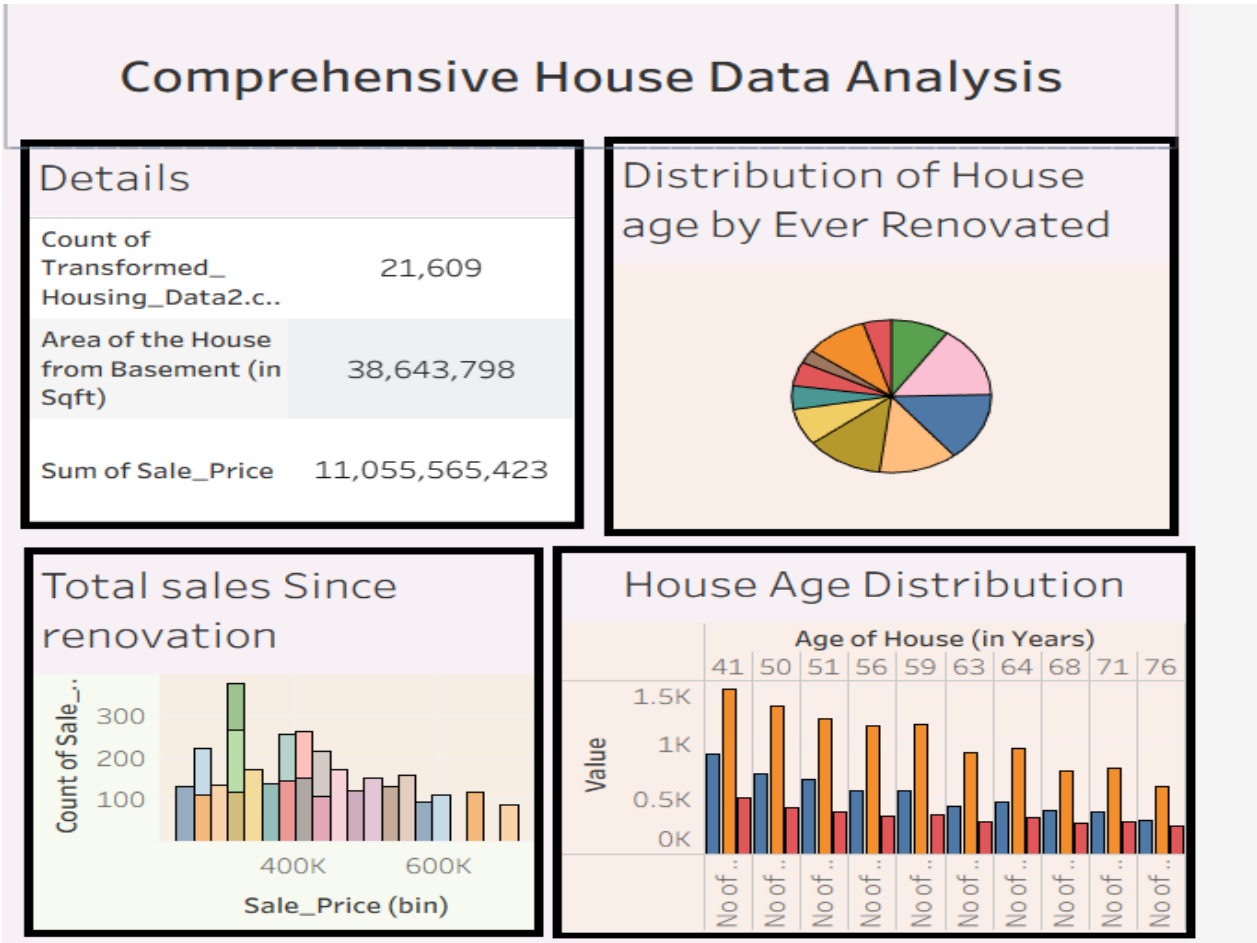


Insert Screenshot 7 – House Age Distribution by Bathrooms, Bedrooms, Floors

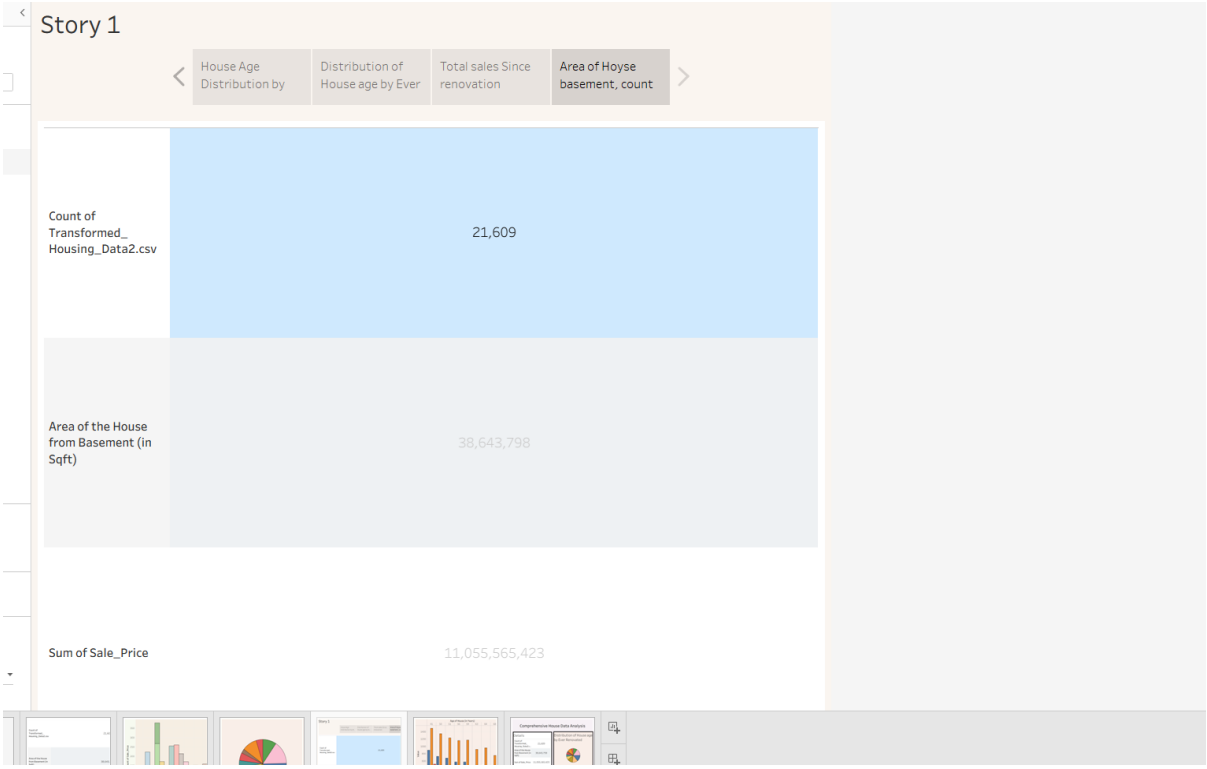


6. Dashboard Design

A Tableau dashboard was created to combine all visualizations into one interactive view. The dashboard allows users to quickly understand housing market patterns and compare different features affecting house prices.



7.Story Board



8.Key Insights

The analysis revealed several insights:

- Houses with larger basement areas generally have higher sale prices.
- Renovated houses tend to have higher market value.
- Houses with more bedrooms and bathrooms often sell for higher prices.
- House age also impacts the price distribution in the market.

9.Tools Used

- Tableau Desktop

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- Kaggle Dataset
- Data Visualization Techniques

10.Conclusion

This Tableau project successfully visualizes housing market trends using interactive dashboards. The analysis helps understand the relationship between house features and sale prices, enabling better insights into housing market behavior.

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